

1. Create user with name Techie and provide sudo access to user.

```
chundurur@chundurur-virtual-machine:~$ sudo adduser techie
Adding user `techie' ...
Adding new group `techie' (1001) ...
Adding new user `techie' (1001) with group `techie' ...
Creating home directory `/home/techie' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for techie
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] Y
chundurur@chundurur-virtual-machine:~$ sudo whoami
root
chundurur@chundurur-virtual-machine:~$ sudo usermod -aG sudo techie
chundurur@chundurur-virtual-machine:~$ group techie
Command 'group' not found, did you mean:
  command 'grop' from deb grop (2:0.10-1.2)
  command 'groups' from deb coreutils (8.32-4.1ubuntu1.3)
Try: sudo apt install <deb name>
chundurur@chundurur-virtual-machine:~$ groups techie
techie : techie sudo
chundurur@chundurur-virtual-machine:~$ su -techie
su: invalid option -- 't'
Try 'su --help' for more information.
chundurur@chundurur-virtual-machine:~$ sudo whoami
root
chundurur@chundurur-virtual-machine:~$
```

- **Purpose:** To learn how to create a Linux user and give them administrative access.
- **Commands & Verify:** `sudo adduser techie`, `sudo usermod -aG sudo techie`, and `groups techie` to check the user's permissions.
- **Conclusion:** The techie user was successfully created and given **sudo (administrator) privileges**.

2. Navigate to the home directory.

```
techie@chundurur-virtual-machine:~$ cd ~
techie@chundurur-virtual-machine:~$ pwd
/home/techie
techie@chundurur-virtual-machine:~$
```

- **Purpose:** To learn how to navigate to the user's home directory and check the current working directory.
- **Commands & Verify:** `cd ~` moves to the home directory, and `pwd` displays the current directory.
- **Conclusion:** The techie user successfully accessed and verified its home directory `/home/techie`.

3. Create a new directory.

```
techie@chundurur-virtual-machine:~$ mkdir karthik
techie@chundurur-virtual-machine:~$ ls
karthik
```

- **Purpose:** To learn how to create a new directory in Linux.

- **Commands & Verify:** `mkdir karthik` creates the directory, and `ls` verifies that it was created.
- **Conclusion:** The `karthik` directory was successfully created in the home directory.

4. List the contents of a directory.

```
chunduru@chunduru-virtual-machine:~$ ls -la
total 100
drwxr-xr-x 14 chunduru chunduru 4096 Aug  8 11:00 .
drwxr-xr-x  4 root      root      4096 Aug  8 11:10 ..
-rw-r--r--  1 chunduru chunduru 2658 Aug  6 16:25 .bash_history
-rw-r--r--  1 chunduru chunduru 220   Aug  4 11:05 .bash_logout
-rw-r--r--  1 chunduru chunduru 3771 Aug  4 11:05 .bashrc
drwxr-xr-x 11 chunduru chunduru 4096 Aug  6 11:38 .cache
drwxr-xr-x 11 chunduru chunduru 4096 Aug  6 11:38 .config
drwxr-xr-x  2 chunduru chunduru 4096 Aug  6 15:04 Desktop
drwxr-xr-x  2 chunduru chunduru 4096 Aug  4 11:20 Documents
drwxr-xr-x  2 chunduru chunduru 4096 Aug  4 11:20 Downloads
-rw-r--r--  1 chunduru chunduru 12288 Aug  6 16:04 .karthik.swo
-rw-r--r--  1 chunduru chunduru 12288 Aug  6 14:55 .karthik.swp
-rw-r--r--  1 chunduru chunduru 20     Aug  6 15:38 .lessht
drwxr-xr-x  3 chunduru chunduru 4096 Aug  4 11:19 .local
drwxr-xr-x  2 chunduru chunduru 4096 Aug  4 11:20 Music
drwxr-xr-x  2 chunduru chunduru 4096 Aug  4 11:20 Pictures
-rw-r--r--  1 chunduru chunduru 807   Aug  4 11:05 .profile
drwxr-xr-x  2 chunduru chunduru 4096 Aug  4 11:20 Public
drwxr-xr-x  3 chunduru chunduru 4096 Aug  4 11:19 snap
-rw-r--r--  1 chunduru chunduru 0      Aug  6 15:32 .sudo_as_admin_successful
drwxr-xr-x  2 chunduru chunduru 4096 Aug  4 11:20 Templates
drwxr-xr-x  2 chunduru chunduru 4096 Aug  4 11:20 Videos
chunduru@chunduru-virtual-machine:~$
```

- **Purpose:** To learn how to list the contents of a directory in Linux.
- **Command & Verify:** `ls` displays the directory contents, such as **F1, F2, and F3**.
- **Conclusion:** The directory contents were successfully listed and verified.

5. Change the current directory.

```
chunduru@chunduru-virtual-machine:~$ ls
Desktop Documents Downloads Music Pictures Public snap Templates Videos
chunduru@chunduru-virtual-machine:~$ cd Desktop/
chunduru@chunduru-virtual-machine:~/Desktop$
```

- **Purpose:** To learn how to navigate from the current directory to another directory in Linux.
- **Command & Verify:** `ls` lists the directories, and `cd Desktop/` moves into the **Desktop** directory.
- **Conclusion:** The user successfully navigated to the `/Desktop` directory.

6. Create a new empty file.

```
chunduru@chunduru-virtual-machine:~$ touch file.text
chunduru@chunduru-virtual-machine:~$ ls
Desktop Documents Downloads file.text Music Pictures Public snap Templates Videos
chunduru@chunduru-virtual-machine:~$
```

- **Purpose:** To learn how to create a new empty file in Linux.
- **Command & Verify:** `touch file.text` creates the file, and `ls` verifies that **file.text** exists.
- **Conclusion:** The `file.text` file was successfully created and verified.

7. View the contents of a file.

```
chunduru@chunduru-virtual-machine:~$ cat file.text
a
b
c
d
e
f
f
g
```

- **Purpose:** To learn how to display the contents of a text file in Linux.
- **Command & Verify:** `cat file.text` displays the contents of the file, such as **a**, **b**, **c**, **d**, **e**, **f**, **g**.

Conclusion: The contents of `file.text` were successfully displayed using the `cat` command.

8. Copy a file to another location.

```
chunduru@chunduru-virtual-machine:~$ cp -r karthik /home/chunduru/Desktop/
chunduru@chunduru-virtual-machine:~$ ls
1 Desktop Documents Downloads file.text karthik Music Pictures Public snap Templates Videos
chunduru@chunduru-virtual-machine:~$ cd Desktop/
chunduru@chunduru-virtual-machine:~/Desktop$ ls
karthik
chunduru@chunduru-virtual-machine:~/Desktop$
```

Purpose: To learn how to copy a directory from one location to another in Linux.

Command & Verify: `cp -r karthik /home/chunduru/Desktop/` copies the `karthik` directory, and `ls` verifies it in `Desktop`.

Conclusion: The `karthik` directory was successfully copied to the `Desktop`.

9. Move a file to another location.

```
chunduru@chunduru-virtual-machine:~/Desktop$ mv karthik /home/chunduru
chunduru@chunduru-virtual-machine:~/Desktop$ cd ~
chunduru@chunduru-virtual-machine:~$ pwd
/home/chunduru
chunduru@chunduru-virtual-machine:~$ ls
1 Desktop Documents Downloads file.text karthik Music Pictures Public snap Templates Videos
chunduru@chunduru-virtual-machine:~$
```

Purpose: To learn how to move a directory from one location to another in Linux.

Command & Verify: `mv karthik /home/chunduru` moves the `karthik` directory, and `ls` verifies it in the home directory.

Conclusion: The `karthik` directory was successfully moved to the home directory.

10. Rename a file.

```
chunduru@chunduru-virtual-machine:~$ mv file.text newfile1.text
chunduru@chunduru-virtual-machine:~$ ls
1 Desktop Documents Downloads karthik Music newfile1.text Pictures Public snap Templates Videos
chunduru@chunduru-virtual-machine:~$
```

- **Purpose:** To learn how to rename a file in Linux.
- **Command & Verify:** `mv file.text newfile1.text` renames `file.text` to `newfile1.text`, and `ls` verifies that `newfile1.text` exists in the directory.
- **Conclusion:** The `file.text` file was successfully renamed to `newfile1.text`.

11. Delete a file.

```
chunduru@chunduru-virtual-machine:~$ rm newfile1.text
chunduru@chunduru-virtual-machine:~$ ls
1 Desktop Documents Downloads karthik Music Pictures Public snap Templates Videos
chunduru@chunduru-virtual-machine:~$
```

- **Purpose:** To learn how to delete a file in Linux.
- **Command & Verify:** `rm newfile1.text` removes the file `newfile1.text`, and `ls` verifies that `newfile1.text` no longer exists in the directory.
- **Conclusion:** The `newfile1.text` file was successfully deleted.

12. Grant or revoke permissions on a file or directory.

```
chunduru@chunduru-virtual-machine:~$ chmod 471 file.text
chunduru@chunduru-virtual-machine:~$ ls -l
total 44
-rw-rw-r-- 1 chunduru chunduru 11 Aug  8 11:58 1
drwxr-xr-x 2 chunduru chunduru 4096 Aug  8 12:14 Desktop
drwxr-xr-x 2 chunduru chunduru 4096 Aug  4 11:20 Documents
drwxr-xr-x 2 chunduru chunduru 4096 Aug  4 11:20 Downloads
-r--rwx--x 1 chunduru chunduru  0 Aug  8 12:26 file.text
drwxrwxr-x 2 chunduru chunduru 4096 Aug  8 12:08 karthik
drwxr-xr-x 2 chunduru chunduru 4096 Aug  4 11:20 Music
drwxr-xr-x 2 chunduru chunduru 4096 Aug  4 11:20 Pictures
drwxr-xr-x 2 chunduru chunduru 4096 Aug  4 11:20 Public
drwx----- 3 chunduru chunduru 4096 Aug  4 11:19 snap
drwxr-xr-x 2 chunduru chunduru 4096 Aug  4 11:20 Templates
drwxr-xr-x 2 chunduru chunduru 4096 Aug  4 11:20 Videos
chunduru@chunduru-virtual-machine:~$
```

- **Purpose:** To learn how to grant or revoke permissions on a file or directory in Linux.
- **Command & Verify:** `chmod 471 file.text` changes the permissions of `file.text`, and `ls -l` verifies the updated permissions (`-r--rwx--x`) for the file.
- **Conclusion:** The permissions for `file.text` were successfully updated using the `chmod` command.

13. View the current date and time.

```
chunduru@chunduru-virtual-machine:~$ date
Saturday 08 August 2026 12:32:44 PM IST
chunduru@chunduru-virtual-machine:~$
```

- **Purpose:** To learn how to view the current date and time in Linux.
- **Command & Verify:** `date` displays the system's current date and time (Saturday 08 August 2026 12:32:44 PM IST).
- **Conclusion:** The current date and time were successfully displayed using the `date` command.

14. Check the system uptime.

```
chunduru@chunduru-virtual-machine:~$ uptime
12:42:05 up 1:53, 1 user, load average: 0.00, 0.00, 0.00
chunduru@chunduru-virtual-machine:~$
```

- **Purpose:** To learn how to check the system uptime in Linux.
- **Command & Verify:** `uptime` displays how long the system has been running (1:53), along with the current time (12:42:05), the number of active users (1 user), and the load averages (0.00, 0.00, 0.00).
- **Conclusion:** The system uptime and load average information were successfully displayed using the `uptime` command.

15. View the running processes.

```
chundurur@chundurur-virtual-machine:~$ ps
  PID TTY          TIME CMD
 2326 pts/0    00:00:00 bash
 3788 pts/0    00:00:00 bash
 4056 pts/0    00:00:00 ps
chundurur@chundurur-virtual-machine:~$
```

- **Purpose:** To learn how to view the running processes in Linux.
- **Command & Verify:** ps displays the active processes running in the current shell session, showing process IDs such as 2326, 3788, and 4056 for processes like bash and ps.
- **Conclusion:** The active running processes were successfully viewed using the ps command.

16. Kill a running process.

```
chundurur@chundurur-virtual-machine:~$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root             1  0.0  0.6 167868 13280 ?        Ss   10:48   0:05 /sbin/init auto noprompt splash
root             2  0.0  0.0      0     0 ?        S    10:48   0:00 [kthreadd]
root             3  0.0  0.0      0     0 ?        S    10:48   0:00 [pool_workqueue_release]
root             4  0.0  0.0      0     0 ?        I<   10:48   0:00 [kworker/R-rcu_g]
root             5  0.0  0.0      0     0 ?        I<   10:48   0:00 [kworker/R-rcu_p]
root             6  0.0  0.0      0     0 ?        I<   10:48   0:00 [kworker/R-slub_]
root             7  0.0  0.0      0     0 ?        I<   10:48   0:00 [kworker/R-netns]
root            11  0.0  0.0      0     0 ?        I    10:48   0:00 [kworker/u256:0-ipv6_addrconf]
root            12  0.0  0.0      0     0 ?        I<   10:48   0:00 [kworker/R-mm_pe]
root            13  0.0  0.0      0     0 ?        I    10:48   0:00 [rcu_tasks_kthread]
root            14  0.0  0.0      0     0 ?        I    10:48   0:00 [rcu_tasks_rude_kthread]
root            15  0.0  0.0      0     0 ?        I    10:48   0:00 [rcu_tasks_trace_kthread]
root            16  0.0  0.0      0     0 ?        S    10:48   0:00 [ksoftirqd/0]
root            17  0.0  0.0      0     0 ?        I    10:48   0:01 [rcu_preempt]
root            18  0.0  0.0      0     0 ?        S    10:48   0:00 [migration/0]
root            19  0.0  0.0      0     0 ?        S    10:48   0:00 [idle_inject/0]
root            20  0.0  0.0      0     0 ?        S    10:48   0:00 [cpuhp/0]
root            21  0.0  0.0      0     0 ?        S    10:48   0:00 [cpuhp/1]
root            22  0.0  0.0      0     0 ?        S    10:48   0:00 [idle_inject/1]
root            23  0.0  0.0      0     0 ?        S    10:48   0:00 [migration/1]
root            24  0.0  0.0      0     0 ?        S    10:48   0:00 [ksoftirqd/1]
root            29  0.0  0.0      0     0 ?        S    10:48   0:00 [kdevtmpfs]
root            30  0.0  0.0      0     0 ?        I<   10:48   0:00 [kworker/R-inet_]
root            31  0.0  0.0      0     0 ?        S    10:48   0:00 [kauditd]
root            33  0.0  0.0      0     0 ?        S    10:48   0:00 [khungtaskd]
root            34  0.0  0.0      0     0 ?        S    10:48   0:00 [oom_reaper]
root            36  0.0  0.0      0     0 ?        I<   10:48   0:00 [kworker/R-write]
root            38  0.1  0.0      0     0 ?        S    10:48   0:10 [kcompactd0]
root            39  0.0  0.0      0     0 ?        SN   10:48   0:00 [ksmd]
root            42  0.0  0.0      0     0 ?        SN   10:48   0:00 [khugepaged]
root            43  0.0  0.0      0     0 ?        I<   10:48   0:00 [kworker/R-kinte]
root            44  0.0  0.0      0     0 ?        I<   10:48   0:00 [kworker/R-kbloc]
root            45  0.0  0.0      0     0 ?        I<   10:48   0:00 [kworker/R-blkcg]
root            46  0.0  0.0      0     0 ?        S    10:48   0:00 [irq/9-acpi]
root            47  0.0  0.0      0     0 ?        I<   10:48   0:00 [kworker/R-tpm_d]
root            48  0.0  0.0      0     0 ?        I<   10:48   0:00 [kworker/R-ata_s]
root            49  0.0  0.0      0     0 ?        I<   10:48   0:00 [kworker/R-md]
root            50  0.0  0.0      0     0 ?        I<   10:48   0:00 [kworker/R-md_bi]
root            51  0.0  0.0      0     0 ?        I<   10:48   0:00 [kworker/R-edac-]
root            52  0.0  0.0      0     0 ?        I<   10:48   0:00 [kworker/R-devfr]
```

- **Purpose:** To learn how to view detailed information about all running processes in Linux.
- **Command & Verify:** ps aux lists all active processes across all users with detailed status information (such as PID, %CPU, %MEM, and COMMAND).
- **Conclusion:** The detailed list of running processes was successfully displayed using the ps aux command.

17. Install a package using the package manager (e.g., apt or yum).

```
chundurur@chundurur-virtual-machine:~$ sudo apt install tree
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcb-1.0-0 libsbios-c2
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  tree
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 47.9 kB of archives.
After this operation, 116 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 tree amd64 2.0.2-1 [47.9 kB]
Fetched 47.9 kB in 1s (66.1 kB/s)
Selecting previously unselected package tree.
(Reading database ... 203578 files and directories currently installed.)
Preparing to unpack .../tree_2.0.2-1_amd64.deb ...
Unpacking tree (2.0.2-1) ...
Setting up tree (2.0.2-1) ...
Processing triggers for man-db (2.10.2-1) ...
chundurur@chundurur-virtual-machine:~$ tree --version
tree v2.0.2 (c) 1996 - 2022 by Steve Baker, Thomas Moore, Francisc Rocher, Florian Sesser, Kyosuke Tokoro
```

- **Purpose:** To learn how to install a software package in Linux using a package manager.
- **Command & Verify:** `sudo apt install tree` installs the tree package using the apt package manager, and running `tree --version` verifies that tree v2.0.2 was successfully installed.
- **Conclusion:** The tree package was successfully installed and verified using the apt package manager.

18. Update the system packages.

```
chundurur@chundurur-virtual-machine:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
chundurur@chundurur-virtual-machine:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcb-1.0-0 libsbios-c2
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
chundurur@chundurur-virtual-machine:~$
```

- **Purpose:** To learn how to update system packages in Linux.
- **Command & Verify:** `sudo apt update` updates the package list from repositories, and `sudo apt upgrade` upgrades all installed software packages to their latest versions, verifying that all packages are up to date (0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded).
- **Conclusion:** The system packages were successfully updated and verified using `sudo apt update` and `sudo apt upgrade`.

19. Create a symbolic link.

```
chunduru@chunduru-virtual-machine:~$ ln -s file.text linl.text
chunduru@chunduru-virtual-machine:~$ ls -l
total 44
-rw-rw-r-- 1 chunduru chunduru 11 Aug  8 11:58 1
drwxr-xr-x 2 chunduru chunduru 4096 Aug  8 12:14 Desktop
drwxr-xr-x 2 chunduru chunduru 4096 Aug  4 11:20 Documents
drwxr-xr-x 2 chunduru chunduru 4096 Aug  4 11:20 Downloads
-r--rwx-- 1 chunduru chunduru  0 Aug  8 12:26 file.text
drwxrwxr-x 2 chunduru chunduru 4096 Aug  8 12:08 karthik
lrwxrwxrwx 1 chunduru chunduru  9 Aug  8 12:53 linl.text -> file.text
drwxr-xr-x 2 chunduru chunduru 4096 Aug  4 11:20 Music
drwxr-xr-x 2 chunduru chunduru 4096 Aug  4 11:20 Pictures
drwxr-xr-x 2 chunduru chunduru 4096 Aug  4 11:20 Public
drwx----- 3 chunduru chunduru 4096 Aug  4 11:19 snap
drwxr-xr-x 2 chunduru chunduru 4096 Aug  4 11:20 Templates
drwxr-xr-x 2 chunduru chunduru 4096 Aug  4 11:20 Videos
chunduru@chunduru-virtual-machine:~$ S
```

- **Purpose:** To learn how to create a symbolic link (symlink) to a file in Linux.
- **Command & Verify:** `ln -s file.text linl.text` creates a symbolic link named `linl.text` pointing to `file.text`, and running `ls -l` verifies that `linl.text -> file.text` is successfully created.
- **Conclusion:** The symbolic link `linl.text` pointing to `file.text` was successfully created and verified.

20. Search for files using the find command.

```
chunduru@chunduru-virtual-machine:~$ find -name "file.text"
./file.text
chunduru@chunduru-virtual-machine:~$
```

- **Purpose:** To learn how to search for files using the `find` command in Linux.
- **Command & Verify:** `find -name "file.text"` searches for a file named `file.text` in the current directory and subdirectories, successfully locating and displaying `./file.text`.
- **Conclusion:** The file `file.text` was successfully found using the `find` command.